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A thermodynamic modeling of the Gd–Tl system

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ABSTRACT

The phase equilibria and thermodynamic properties of the Gd–Tl system were optimized using the CALPHAD (CALculation of PHAse Diagram) technique. In the present work, the excess Gibbs energies of the liquid, bcc and hcp solution phases were modeled using the polynomial temperature dependence and Kaptay equation. The intermetallic compounds Gd_2Tl, Gd_3Tl_5 and $GdTl_3$ were treated as stoichiometric compounds. The intermetallic compound Gd_5Tl_3, which had a homogeneity range, was treated with a sublattice model $(Gd,Tl)_5Tl_3$. A sublattice model $(Gd,Tl)_{0.5}(Gd,Tl)_{0.5}$ was applied to describe the compound GdTl in order to cope with the order–disorder transition between GdTl with CsCl-type structure (B2) and body-centered cubic solution (A2) in the Gd–Tl system. Two sets of self-consistent thermodynamic parameters were obtained.

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1. Introduction

Gd and Tl elements are the important alloying element in the thermoelectric material $FeSb_2$ for thermoelectric purposes at a temperature of 150 K or less, in which Gd can substitute all or part of the Fe atoms and Tl can substitute all or part of the Sb atoms. The Gd–Tl system is one of the binary systems studied as the thermoelectric material at a temperature of 150 K or less [1].

The available experimental crystallographic data and phase equilibria data of the Gd–Tl binary system in the literature up to 1990 had been critically reviewed by Delfino et al. [2]. In the assessed phase diagram, there are three solutions liquid, bcc and hcp and six intermetallic compounds Gd_2Tl, Gd_5Tl_3, Gd_5Tl_{3+x}, GdTl, Gd_3Tl_5, and $GdTl_3$ were included. The phase equilibria in the Gd–Tl system were first determined by Delfino et al. [3,4]. There were three disordered solution phases liquid, body-centered cubic (bcc) and hexagonal close packed (hcp) as well as five intermetallic compounds Gd_2Tl, Gd_5Tl_3, GdTl, Gd_3Tl_5 and $GdTl_3$, in which GdTl and $GdTl_3$ melt congruently, and the other compounds were formed by peritectic reactions. Saccone et al. [5] re-measured the phase equilibria between 36 and 47 at.% Tl in the Gd–Tl system. The compound Gd_5Tl_3 was determined to melt congruently, and a new compound Gd_5Tl_{3+x} with the Ba_5Pb_3-type structure was reported. Due to a lack of the experimental data, the compound Gd_5Tl_{3+x} was not considered in the present work. The invariant reaction liq. → Gd_5Tl_{3+x} + GdTl assumed by Saccone et al. [5] was substituted by liq. → Gd_5Tl_3 + GdTl reported by Delfino et al. [4] in the present work. In addition, the crystal structures of the compounds in the Gd–Tl system were studied by many researchers [6–10], and were listed in Table 1.

The enthalpy of formation of $GdTl_3$ was determined by Palenzona and Cirafici [11] using dynamic differential calorimetry and Demykina et al. [12] using EMF method. The experimental results are −36.8 kJ/mol [11] and −36.4 kJ/mol of atoms [12], respectively. In the present work, the enthalpies of formation of compounds Gd_2Tl, Gd_5Tl_3, GdTl, Gd_3Tl_5, and $GdTl_3$ at 298 K were predicted using revised Miedema model [13], in kJ/mol of atoms: Gd_2Tl −54.4, Gd_5Tl_3 −59.2, GdTl −65.8, Gd_3Tl_5 −59.05 and $GdTl_3$ −42.06.

This work deals with an assessment of the thermodynamic description of the Gd–Tl system by means of the CALPHAD (CALculation of PHAse Diagram) technique.

2. Thermodynamic models

2.1. Unary phases

The Gibbs energy function $G_i^{\phi}(T) = {}^0G_i^{\phi}(T) - H_i^{SER}(298.15\,K)$ for the element i (i = Gd, Tl) in the phase ϕ (ϕ = liquid, body-centered cubic (bcc), or hexagonal close packed (hcp)) is described by an equation of the following form:

$$G_i^{\phi}(T) = a + bT + cT\ \ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where $H_i^{SER}(298.15\,K)$ is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, hcp for Gd and Tl. The Gibbs energy of the element i, $G_i^{\phi}(T)$, in its SER state, is denoted by $GHSER_i$, i.e.

$$GHSER_{Gd} = {}^0G_{Gd}^{hcp}(T) - H_{Gd}^{SER}(298.15\,K) \quad (2)$$

$$GHSER_{Tl} = {}^0G_{Tl}^{hcp}(T) - H_{Tl}^{SER}(298.15\,K) \quad (3)$$

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Table 1
The crystal structure of the compounds in the Gd–Tl system.

Phase	Approximate composition range (at.% Tl)	Pearson symbol	Structural designation	Prototype	Reference
Gd_2Tl	32.8–33.8	hP6	$B8_2$	Ni_2In	[9]
Gd_5Tl_3	37.0–38.0	hP16	–	Mn_5Si_3	[8]
GdTl	48.4–51.1	cP2	B2	CsCl	[7]
Gd_3Tl_5	62.5–63.0	oC32	–	Pd_5Pu_3	[10]
$GdTl_3$	75.0	cP4	$L1_2$	$AuCu_3$	[6,7]

In the present work, the Gibbs energy functions are taken from the SGTE (Scientific Group Thermodata Europe) pure elements database compiled by Dinsdale [14] and are listed in Table 2.

2.2. Solution phases

In the Gd–Tl system, there are three solution phases: liquid, bcc and hcp. Their Gibbs energies are described by the following expression:

$$G_m^{\phi} = x_{Gd}G_{Gd}^{\phi}(T) + x_{Tl}G_{Tl}^{\phi}(T) + RT(x_{Gd}\ \ln x_{Gd} + x_{Tl}\ln x_{Tl}) + {}^{E}G_m^{\phi} + {}^{mg}G_m^{\phi} \quad (4)$$

where $G_i^{\phi}(T)$ is the Gibbs energy for the element i (i = Gd, Tl) in the phase ϕ (ϕ = liquid, bcc and hcp), and is taken from the SGTE (Scientific Group Thermodata Europe) pure elements database [14]; ${}^{mg}G_m^{\phi}$ is the magnetic contribution to the Gibbs energy. ${}^{E}G_m^{\phi}$ is the excess Gibbs energy, expressed by the Redlich–Kister polynomial [15]:

$${}^{E}G_m^{\phi} = x_{Gd}x_{Tl}\sum_j {}^{j}L^{\phi}(x_{Gd} - x_{Tl})^j \quad (5)$$

where ${}^{j}L^{\phi}$ is the interaction parameter between elements Gd and Tl, which is to be evaluated in the present work. Its general form is

$${}^{j}L^{\phi} = a_j + b_jT + c_jT\ \ln T + d_jT^2 + e_jT^3 + f_jT^{-1} \quad (6)$$

The polynomial temperature dependence of the interaction parameters in the Redlich–Kister equation, described by Eq. (6) does not automatically correspond to reasonable boundary conditions. The thermodynamic description may become valid over a wide temperature range, but there always is a possible temperature at which the parameter ${}^{j}L_{Gd,Tl}^{liq.}$ changes sign. When the change is from negative to positive, the excess Gibbs energy will change its sign at a certain high temperature and will have an infinite value at infinitely high temperature, which is in conflict with thermodynamic constraints and spurious miscibility gaps tend to appear. These miscibility gaps do not close and imply that the solution is unstable. To solve this problem, Kaptay [16] has proposed a semi-empirical equation:

$${}^{j}L^{\phi} = A\ \exp\left(-\frac{B}{A}T\right) \quad (7)$$

where A and B represent the enthalpic and entropic contributions to the excess Gibbs energy at 0 K. Because of the same sign for A and B, the expression in Eq. (7) indicates that there is a finite contribution to the excess Gibbs energy of a phase at high temperature. When the temperature increases, the interactions weaken, making the solution ideal as $T \rightarrow \propto$, which can avoid the appearance of the miscibility gap.

In the present work, two sets of the interaction parameters for the liquid and solid solutions (bcc, hcp) are optimized using the polynomial temperature dependence described by Eq. (6) and the exponential temperature dependence proposed by Kaptay equation [16].

2.3. Stoichiometric intermetallic compounds Gd_2Tl, Gd_3Tl_5 and $GdTl_3$

The intermetallic compounds Gd_2Tl, Gd_3Tl_5 and $GdTl_3$, which have a homogeneity range about 1 at.%, are treated as stoichiometric compounds in the present work.

The Gibbs energy per mole of formula unit Gd_mTl_n is expressed as following:

$$G_m^{Gd_mTl_n} = m\mathrm{GHSER}_{Gd} + n\mathrm{GHSER}_{Tl} + \Delta G_f^{Gd_mTl_n} \quad (8)$$

where $\Delta G_f^{Gd_mTl_n}$ is the Gibbs energy of formation per mole of formula unit Gd_mTl_n. Owing to a lack of experimental data, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e. $\Delta Cp_f = 0$. Thus $\Delta G_f^{Gd_mTl_n}$ can be given by the following expression:

$$\Delta G_f^{Gd_mTl_n} = a + bT \quad (9)$$

where the parameters a and b were evaluated in the present work.

2.4. Intermetallic compounds Gd_5Tl_3

At about 37.5 at.% Tl in the Gd–Tl phase diagram reported by Saccone et al. [5], there were two compounds Gd_5Tl_3 with Mn_5Si_3-type and Gd_5Tl_{3+x} with Ba_5Pb_3-type structures. But the formation temperature of the compound Gd_5Tl_{3+x} with the Ba_5Pb_3-type structure was not determined. In the present work, two compounds Gd_5Tl_3 and Gd_5Tl_{3+x} were treated as one compound with the formula $(Gd,Tl)_5Tl_3$. The Gibbs energy per mole of formula unit Gd_5Tl_3 is given by the following expression:

$$G_m^{Gd_5Tl_3} = y_{Gd}G_{Gd:Tl}^{Gd_5Tl_3} + y_{Tl}G_{Tl:Tl}^{Gd_5Tl_3} + RT(y_{Gd}\ \ln y_{Gd} + y_{Tl}\ln y_{Tl}) + y_{Gd}y_{Tl}\sum_j {}^{j}L_{Gd,Tl:Tl}^{Gd_5Tl_3}(y_{Gd} - y_{Tl})^j \quad (10)$$

where y_{Gd} and y_{Tl} are the site fraction of Gd and Tl on the first sublattice, respectively. The two parameters denoted $G_{Gd:Tl}^{Gd_5Tl_3}$ and $G_{Tl:Tl}^{Gd_5Tl_3}$ represent the Gibbs energies of the compound Gd_5Tl_3 when the first sublattice is occupied by only one element Gd or Tl (y_{Gd} = 1 or y_{Tl} = 1), respectively, which are relative to the enthalpies of pure hcp for Gd and Tl in their SER states. ${}^{j}L_{Gd,Tl:Tl}^{Gd_5Tl_3}$ represents the jth interaction parameters between the elements Gd and Tl on the first sublattice.

2.5. Intermetallic compounds GdTl

The intermetallic compound GdTl has a homogeneity range of about 2 at.% on the Gd-rich side and 2 at.% on the Tl-rich side [4], respectively. GdTl has a B2-type structure and is isotypic with CsCl, in which Gd occupies 1a position, and Tl occupies 1b position. When the Gd atoms in the sublattice are completely substituted by Tl, the structure becomes the disordered bcc-A2 structure, vice versa. The order–disordered transition between bcc-A2 and GdTl-B2 is assumed in the present work. In order to reflect the structural relationship between the compound GdTl (ordered, bcc-B2) and the bcc terminal solution phase (disordered, bcc-A2), in the present work, the two phases were modeled using a single sublattice formalism,

Table 2
Thermodynamic parameters in the Gd–Tl system[a].

Phase	Thermodynamic parameters		Reference
Liquid: $(Gd,Tl)_1$	G(liquid,Gd)=		[14]
	298–1000	$+13060.0262 - 8.32179974T + GHSER_{Gd}$	
	1000–1508	$+146262.037 - 1208.70685T + 159.352082T\ln T - 0.108247135T^2 + 1.06945505\times10^{-5}T^3 - 19678357T^{-1}$	
	1508–3600	$-5397.314 + 192.336215T - 38.5075T\ln T$	
	G(liquid,Tl)=		[14]
	200–577	$-946.623 + 0.755649T - 7.44455T\ln T - 0.044350292T^2 + 1.4248046\times10^{-5}T^3 - 54228T^{-1}$	
	577–1800	$-614.74 + 98.472609T - 25.8437T\ln T - 8.3662\times10^{-4}T^2 + 9\times10^{-12}T^3 - 612570T^{-1}$	
	${}^0L^{liq.}_{Gd,Tl} = -146864.3 + 21.6613T$		This work
	${}^1L^{liq.}_{Gd,Tl} = +2734.5 - 2.1800T$		This work
	${}^2L^{liq.}_{Gd,Tl} = +24031.5$		This work
	$GHSER_{Gd}$=		[14]
	200–1000	$-6834.5855 + 97.13101T - 24.7214131T\ln T - 0.002852405 21T^2 - 3.14674076\times10^{-7}T^3 - 8665.73348T^{-1}$	
	1000–1508	$-6483.25362 + 95.6919924T - 24.6598297T\ln T - 0.00185225011T^2 - 6.61211607\times10^{-7}T^3$	
	1508–3600	$-123124.992 + 699.125537T - 101.800197T\ln T - 0.0150644246T^2 + 6.39165948\times10^{-7}T^3 + 29356890.3T^{-1}$	
	$GHSER_{Tl}$=		[14]
	200–577	$-8104.038 + 107.140405T - 25.2274T\ln T - 0.0033063T^2 - 1.21807\times10^{-7}T^3 + 42058T^{-1}$	
	577–1800	$-15406.859 + 196.771926T - 38.4130658T\ln T + 0.005228106T^2 - 5.19136\times10^{-7}T^3 + 729665T^{-1}$	
bcc (disordered part of B2): $(Gd,Tl)_1(Va)_3$	G(bcc,Gd:Va;0)=		[14]
	200–1000	$+3233.80866 - 2.11184585T + 1.00000001\times10^{-7}T\ln T + GHSER_{Gd}$	
	1000–1508	$+152792.743 - 1349.58873T + 180.097094T\ln T - 0.119550229T^2 + 1.17915728\times10^{-5}T^3 - 22038836T^{-1}$	
	1508–1586	$-15783.7618 + 202.222057T - 38.960425T\ln T$	
	1586–3600	$-19850.5562 + 224.817909T - 41.904333T\ln T + 8.58222759\times10^{-4}T^2 - 3.77570269\times10^{-8}T^3 + 995428.573T^{-1}$	
	G(bcc,Tl:Va;0)=		[14]
	298–577	$-9194.493 + 150.019517T - 33.0508T\ln T + 0.0172318T^2 - 1.0115933\times10^{-5}T^3 + 82153T^{-1}$	
	577–1800	$-41836.403 + 482.633817T - 79.2926704T\ln T + 0.026042993T^2 - 2.359101\times10^{-6}T^3 + 3507810T^{-1}$	
	${}^0L^{bcc}_{Gd,Tl} = -122898.9 + 16.3141T$		This work
	${}^1L^{bcc}_{Gd,Tl} = -3682.3 - 4.8594T$		This work
	${}^2L^{bcc}_{Gd,Tl} = +21842.4$		This work
hcp: $(Gd,Tl)_1(Va)_1$	G(hcp,Gd:Va;0) = $+GHSER_{Gd}$		[14]
	G(hcp,Tl:Va;0) = $+GHSER_{Tl}$		[14]
	${}^0L^{hcp}_{Gd,Tl} = -93688.3 + 18.6685T$		This work
	${}^1L^{hcp}_{Gd,Tl} = -18114.4 + 3.2990T$		This work
	${}^2L^{hcp}_{Gd,Tl} = -10910.5$		This work
GdTl (ordered part of B2): $(Gd,Tl)_{0.5}(Gd,Tl)_{0.5}$	$G^{GdTl}_{Gd:Tl} = G^{GdTl}_{Tl:Gd} = -38419.3 + 2.5000T$		This work
	${}^0L^{GdTl}_{Gd,Tl:Gd} = {}^0L^{GdTl}_{Gd:Gd,Tl} = -47247.9 - 16.9662T$		This work
	${}^1L^{GdTl}_{Gd,Tl:Gd} = {}^1L^{GdTl}_{Gd:Gd,Tl} = -31000.0$		This work
	${}^0L^{GdTl}_{Gd,Tl:Tl} = {}^0L^{GdTl}_{Tl:Gd,Tl} = +42598.5 + 13.2217T$		This work
	${}^1L^{GdTl}_{Gd,Tl:Tl} = {}^1L^{GdTl}_{Tl:Gd,Tl} = -15419.6$		This work
Gd_2Tl	$G^{Gd_2Tl}_{Gd:Tl} = 2GHSER_{Gd} + GHSER_{Tl} - 113776.3 + 8.7345T$		This work
Gd_5Tl_3: $(Gd,Tl)_5Tl_3$	$G^{Gd_5Tl_3}_{Gd:Tl} = 5GHSER_{Gd} + 3GHSER_{Tl} - 322933.6 + 17.7346T$		This work
	$G^{Gd_5Tl_3}_{Tl:Tl} = +8GHSER_{Tl} + 40000.0$		This work
	${}^0L^{Gd_5Tl_3}_{Gd,Tl:Tl} = -311208.5$		This work
Gd_3Tl_5	$G^{Gd_3Tl_5}_{Gd:Tl} = 3GHSER_{Gd} + 5GHSER_{Tl} - 345875.5 + 33.4500T$		This work
$GdTl_3$	$G^{GdTl_3}_{Gd:Tl} = GHSER_{Gd} + 3GHSER_{Tl} - 146770.6 + 18.6130T$		This work

[a] In SI units (Joule, mole of the formula units and Kelvin).

represented by $(Gd,Tl)_{0.5}$:$(Gd,Tl)_{0.5}$. The Gibbs energy per mole of formula unit GdTl is given by the following expression:

$$G_m^{GdTl} = G_m^{dis}(x_i) + G_m^{ord}(y_i', y_i'') - G_m^{ord}(x_i) \quad (11)$$

where x_i is the mole fraction of Gd or Tl; y_i' and y_i'' are the site fractions of Gd or Tl on the first and second sublattices, respectively; x_i is related with y_i' and y_i'' according to $x_i = 0.5y_i' + 0.5y_i''$; $G_m^{dis}(x_i)$ is molar Gibbs energy of the disordered bcc phase and the expression is same to Eq. (4); $G_m^{ord}(y_i', y_i'') - G_m^{ord}(x_i)$ is ordering contributions. $G_m^{ord}(y_i', y_i'')$ is calculated using Eq. (12) below, then $G_m^{ord}(x_i)$ is calculated using the same model parameters but replacing y_i' and y_i'' with mole fraction x_i in Eq. (12). When the phase becomes disor-

dered, i.e. $y'_i = y''_i = x_i$, it can be seen that the contributions from the ordering, i.e. the second and third terms in Eq. (11), will cancel and only the first term $G_m^{dis}(x_i)$ will give a contribution to the Gibbs energy.

$$\begin{aligned} G_m^{ord}(y'_i, y''_i) &= y'_{Gd}y''_{Tl}G^{GdTl}_{Gd:Tl} + y'_{Gd}y''_{Tl}G^{GdTl}_{Gd:Tl} + y'_{Gd}y''_{Tl}G^{GdTl}_{Gd:Tl} \\ &+ y'_{Gd}y''_{Tl}G^{GdTl}_{Tl:Tl} + 0.5RT(y'_{Gd}\ \ln y'_{Gd} + y'_{Tl}\ln y'_{Tl}) \\ &+ 0.5RT(y''_{Gd}\ln y''_{Gd} + y''_{Tl}\ln y''_{Tl}) + y'_{Gd}y'_{Tl} \\ &\times \left(y''_{Gd}\sum_j {}^jL^{GdTl}_{Gd,Tl:Gd}(y'_{Gd} - y'_{Tl})^j + y''_{Tl}\sum_j {}^jL^{GdTl}_{Gd,Tl:Tl}(y'_{Gd} - y'_{Tl})^j \right) \\ &+ y''_{Gd}y''_{Tl}\left(y'_{Gd}\sum_j {}^jL^{GdTl}_{Gd:Gd,Tl}(y''_{Gd} - y''_{Tl})^j + y'_{Tl}\sum_j {}^jL^{GdTl}_{Tl:Gd,Tl}(y''_{Gd} - y''_{Tl})^j \right) \end{aligned} \quad (12)$$

where y'_i and y''_i are the site fractions of Gd or Tl on the first and second sublattices, respectively; the parameter $G^{GdTl}_{*:*}$ represents the Gibbs energy of the compound GdTl when the first and second sublattices are occupied by only one element Gd or Tl, respectively. ${}^jL^{GdTl}_{Gd,Tl:*}$ and ${}^jL^{GdTl}_{*:Gd,Tl}$ represent the jth interaction parameters between the elements Gd and Tl on the first and second sublattices, respectively.

3. Optimizing procedure

In the present work, the phase relations and transformation temperatures are based on the phase diagram of the Gd–Tl system determined by Delfino et al. [4] and Saccone et al. [5]. The enthalpy of formation of the compound $GdTl_3$ at 298 K were determined by Palenzona and Cirafici [11] and Demykina et al. [12]. Because the enthalpy of formation −42.06 kJ/mol of atoms for $GdTl_3$ predicted by the revised Miedema model is more exothermic than the experimental results −36.8 and −36.4 kJ/mol of atoms [11,12], in the present work, the experimental data determined by Palenzona and Cirafici [11] and Demykina et al. [12] are given a larger weight.

The optimization was carried out by means of the Thermo-Calc software [17], which can handle various kinds of experimental data. This module works by minimizing the square sum of the differences between the experimental data and calculated values. During the optimization, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits.

The optimization was carried out in two steps. In the first treatment, Gd_5Tl_3 an GdTl were assumed to be stoichiometric compounds; in the second treatment, the compound Gd_5Tl_3 was treated as the formula $(Gd,Tl)_5Tl_3$ by using a two-sublattice model, which are described in Section 2.4, and the compound GdTl with CsCl-type structure was treated as the formula $(Gd,Tl)_{0.5}(Gd,Tl)_{0.5}$ and the order–disorder transition between the disordered bcc-A2 and ordered GdTl-B2 was considered. The parameters obtained from the first treatment were used as starting values for the second treatment.

In addition, the Kaptay equation [16] was applied in the solutions liquid, bcc and hcp in order to avoid the metastable miscibility gap at high temperature in the liquid phase.

For the liquid, bcc and hcp solutions, the coefficients ${}^0L^{liq.}$, ${}^1L^{liq.}$ and ${}^2L^{liq.}$, ${}^0L^{bcc}$, ${}^1L^{bcc}$ and ${}^2L^{bcc}$, ${}^0L^{hcp}$, ${}^1L^{hcp}$ and ${}^2L^{hcp}$ in Eq. (4) can be reliably obtained from the experimental data.

The GdTl has a B2-type structure and is isotypic with CsCl. The Gibbs energy of the phase GdTl is assumed to be the sum of the Gibbs energy of the disordered bcc phase and the Gibbs energy needed to form the ordered GdTl from the disordered bcc phase. To obtain the total Gibbs energy of the phase, the disordered part is used as the reference for the Gibbs energy contribution due to ordering, implying that $G^{GdTl}_{Gd:Gd}$ and $G^{GdTl}_{Tl:Tl}$ are zero. For the reason of crystallographic symmetry, we have the following parameter constraints [18,19]:

$$G^{GdTl}_{Gd:Tl} = G^{GdTl}_{Tl:Gd} \quad (13)$$

$${}^jL^{GdTl}_{Gd,Tl:Gd} = {}^jL^{GdTl}_{Gd:Gd,Tl} \quad (14)$$

$${}^jL^{GdTl}_{Tl:Gd,Tl} = {}^jL^{GdTl}_{Gd,Tl:Tl} \quad (15)$$

4. Results and discussion

A thermodynamic description of the Gd–Tl system obtained in the present work is listed in Table 2. The interaction parameters of liquid, bcc and hcp solutions modeled with Kaptay equation [16] are listed in Table 3. The calculated temperatures, compositions for the invariant reactions in the Gd–Tl system using polynomial temperature dependence and Kaptay equation [16] are listed in Table 4. As shown in this table, satisfactory agreement is obtained between the calculations and experimental values [4,5].

Fig. 1a and b is the calculated Gd–Tl phase diagram by means of the thermodynamic parameters using the polynomial temperature dependence and Kaptay equation [16], respectively, and identical to the experimental phase diagram reported by Delfino et al. [4] and Saccone et al. [5].

The calculated enthalpies of formation at 298 K and comparison with the experimental data [11,12] and data calculated by the revised Miedemia model [13] is shown in Fig. 2. The reference states are hcp for two elements Gd and Tl. Satisfactory agreement is obtained between the calculated results and experimental data [11,12].

Fig. 3 is the calculated molar Gibbs energies of the bcc solution (A2) and GdTl with CsCl-type structure (B2) at 1300 K in the Gd–Tl system using polynomial temperature dependence and Kaptay equation [16]. The reference states are bcc for two elements Gd and Tl.

According to the description in Section 2.2, the interaction parameters expressed using Kaptay equation in the excess Gibbs energy for solution phases have certain physical interest, and can avoid the artificial miscibility gap in liquid at elevated temperature. Usually, the thermodynamic parameters optimized using

Table 3
Thermodynamic parameters of the liquid, bcc and hcp solution phases using the Kaptay equation in the Gd–Tl system[a].

Phase	Parameters	Reference
Liquid	Model: $(Gd,Tl)_1$	
	${}^0L^{liq.}_{Gd,Tl} = -150166.0\exp(-1.8089\times10^{-4}T)$	This work
	${}^1L^{liq.}_{Gd,Tl} = +1147.4\exp(-4.4514\times10^{-3}T)$	This work
	${}^2L^{liq.}_{Gd,Tl} = +23822.8$	This work
bcc	Model: $(Gd,Tl)_1(Va)_3$	
	${}^0L^{bcc}_{Gd,Tl} = -124432.4\ \exp(-1.5528\times10^{-4}T)$	This work
	${}^1L^{bcc}_{Gd,Tl} = -5271.8$	This work
	${}^2L^{bcc}_{Gd,Tl} = +16003.3$	This work
hcp	Model: $(Gd,Tl)_1(Va)_1$	
	${}^0L^{hcp}_{Gd,Tl} = -91536.2\ \exp(-2.1808\times10^{-4}T)$	This work
	${}^1L^{liq.}_{Gd,Tl} = -28057.4\exp(-2.1166\times10^{-4}T)$	This work
	${}^2L^{liq.}_{Gd,Tl} = -4866.3$	This work

[a] In SI units (Joule, mole of the formula units and Kelvin).

Table 4
Invariant reactions in the Gd–Tl system.

Reaction	Polynomial				Kaptay equation [16]				Experimental data [4,5]			
	T (K)		x(Tl)		T (K)		x(Tl)		T (K)		x(Tl)	
bcc → hcp + Gd_2Tl	1143	0.0717	0.1300		1143	0.0755	0.1281		1143	0.053	0.137	
liq. → bcc + Gd_2Tl	1193	0.2047	0.1393		1193	0.2053	0.1363		1193	0.230	0.145	
liq. + Gd_5Tl_3 → Gd_2Tl	1328	0.2666	0.3752		1328	0.2665	0.3752		1328	0.263	0.375	
liq. → Gd_5Tl_3	1493	0.3842	0.3842		1493	0.3841	0.3841		1493	0.375	0.375	
liq. → Gd_5Tl_3 + GdTl	1473	0.4255	0.3956	0.4852	1473	0.4262	0.3956	0.4880	1473	0.400		
liq. → GdTl	1543	0.4977	0.4977		1543	0.4984	0.4984		1543	0.500	0.500	
liq. + GdTl → Gd_3Tl_5	1323	0.6253	0.5116		1322	0.6265	0.5145		1323	0.625	0.515	
liq. → Gd_3Tl_5 + $GdTl_3$	1243	0.7069			1243	0.7071			1243	0.715		
liq. → $GdTl_3$	1263	0.750			1263	0.750			1263	0.750		
liq. → $GdTl_3$ + bcc	576	0.9988		1.0000	576	0.9989		0.9999	576	0.999		1.000
bcc → hcp	507	1.0000		1.0000	507	1.0000		1.0000	507[a]	1.000		1.000

[a] The temperature was determined using the Gibbs energy functions of bcc and hcp from the SGTE compilation of Dinsdale [14].

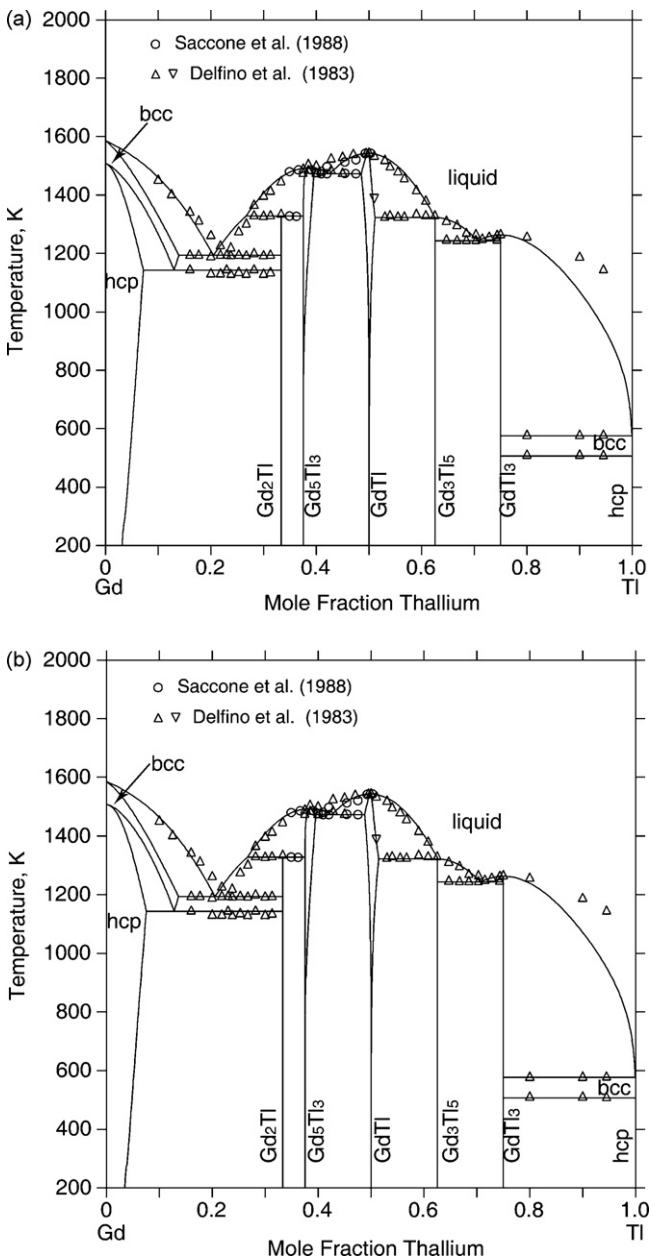

Fig. 1. (a) Calculated Gd–Tl Phase diagram using polynomial temperature dependence with the experimental data [4,5]. (b) Calculated Gd–Tl phase diagram using Kaptay equation [16] with the experimental data [4,5].

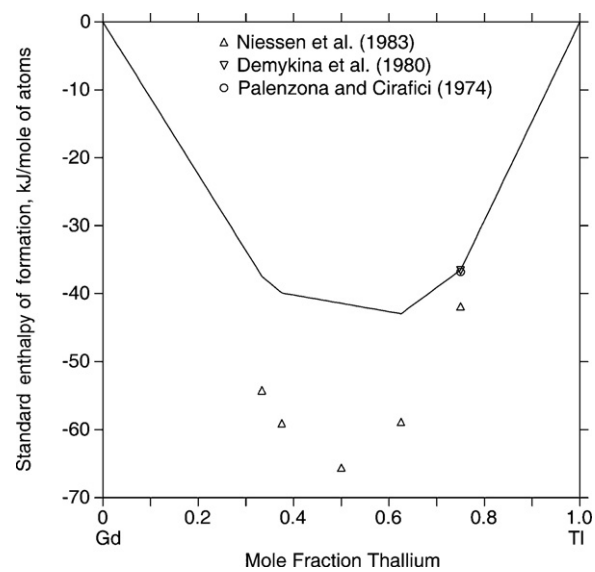

Fig. 2. Calculated standard enthalpy of formation at 298 K in the Gd–Tl system using the present thermodynamic description with the experimental data [11,12] and the predicted values by the revised Miedemia model [13]. The reference states are hcp for Gd and Tl.

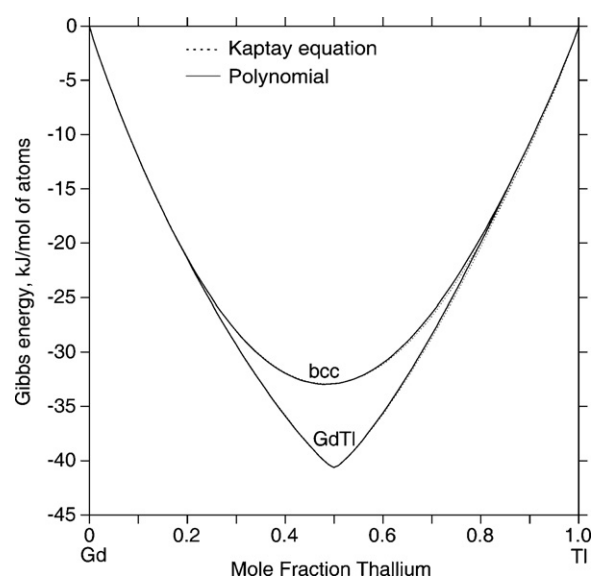

Fig. 3. Calculated molar Gibbs energies of the bcc solution (A2) and GdTl with CsCl-type structure (B2) at 1300 K using polynomial temperature dependence and Kaptay equation [16] in the Gd–Tl system. The reference states are bcc for Gd and Tl.

Kaptay equation model are recommended to choose. However, in past decades, the interaction parameters in the excess Gibbs energy of the solution phases for almost alloy systems were expressed in polynomial dependence of temperature, which were also used in almost thermodynamic databases. In the present work, the thermodynamic parameters using two models for solution phases are optimized.

5. Conclusions

The phase relations and thermodynamic description of the Gd–Tl system were critically evaluated from the experimental information available in the literature. A good agreement is obtained between the experiments and two calculations according to the polynomial temperature dependence and Kaptay equation. With the present optimized parameters, various thermodynamic calculations of practical interest can be made.

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